

JERWIN B. MASAGCA

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Summary

Junior Full-Stack Developer with internship experience working on real-world web applications, backend systems, databases, and authentication. I’ve worked with Node.js, PHP, TypeScript, Django, React, and Supabase, with experience building features, troubleshooting system issues, improving performance, and implementing authentication and access control.

Work Experience

Full-Stack Developer Intern — Microgenesis Business Systems

Sept 2025 - Feb 2026

- Developed backend features using Node.js, TypeScript, and PHP.
- Worked on CodeIgniter modules involving backend logic, data workflows, and system features.
- Contributed to the development of MGEN Central Hub from its early stages, implementing backend features, system integrations, bug fixes, and performance improvements.
- Independently implemented Microsoft Single Sign-On (SSO) using Microsoft Entra ID (Azure Active Directory) for secure authentication.
- Led the migration of the system database to Supabase, ensuring reliable connectivity, data integrity, and proper database handling.
- Designed and implemented Role-Based Access Control (RBAC), allowing Super Admins to manage department and employee access permissions.
- Built and enhanced system features based on requirements and acceptance criteria.
- Worked on the Renewal Care (Support Entitlement) system, implementing system enhancements and fixes.
- Improved system performance by optimizing database queries and limiting heavy data loads to prevent timeouts.
- Supported UI fixes, troubleshooting, and layout improvements when needed.

Project Experience

Dawnasyon: A Smart Relief Distribution System for Barangay Sta. Lucia

Role: Lead Full-Stack Developer

Capstone Project — Quezon City University

- Development: Designed and developed the desktop application independently, handling the frontend, backend, and database integration.
- Authentication: Implemented facial recognition and email-based two-factor authentication (2FA) for administrator access.
- Offline Support: Implemented offline functionality using SQLite for local data persistence, allowing core system functions to continue during limited connectivity and synchronize data when a connection is restored.
- QR Code Integration: Implemented QR code scanning for resident verification and relief goods distribution tracking.
- Mapping: Developed an interactive 3D mapping feature with Street View integration to visualize resident locations and distribution areas.
- Data Management: Developed file import/export features for resident data and automated report generation.
- AI Integration: Integrated “Dawnasyon Assistant,” an AI-powered chatbot using the Gemini API to provide information and support for staff.

Kyusi Esports Community System – Frontend Developer

Quezon City University Organization Project

- Developed the frontend of the community system used by the Kyusi Esports Community.
- Used HTML, CSS, JavaScript, and Bootstrap to build responsive pages for blogs, announcements, events, partners, and media content.
- Implemented consistent layouts and mobile-friendly interfaces for organization members.

Skills

- Languages: JavaScript, TypeScript, PHP, Python, Java
- Backend: Node.js, Django, CodeIgniter
- Frontend: React, Bootstrap, HTML, CSS
- Database: PostgreSQL, MySQL, SQLite, Supabase
- Tools: Git, GitHub, XAMPP, Eclipse
- Other: REST APIs, Responsive UI, Authentication, RBAC

Education

Bachelor of Science in Information Technology (BSIT)

2022 — 2026

Quezon City University| GPA: 1.60

Senior High School

2020 — 2022

International Christian School & Jose Maria Panganiban

Certifications

- JavaScript Essentials 1 — Cisco Networking Academy
- Python Essentials 1 — Cisco Networking Academy
- Networking Basics — Cisco Networking Academy
- Introduction to Cybersecurity — Cisco Networking Academy